

Contiuous Assement I

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Question 1

Throughout this question I make use of Lorentze invariants which I will derieve now. Given the fact that scalars are invariant we can say.

$$F'_{\mu\nu}F^{\mu\nu'} = F_{\mu\nu}F^{\mu\nu} \quad (1)$$

$$= 2F_{0i}F^{0i} + F_{ij}F^{ij} \quad (2)$$

$$= -2 \sum_i E_i E_i + \sum_i \epsilon_{ijk} \epsilon_{ij'k'} F^{jk} F^{j'k'} \quad (3)$$

Given the identity

$$\epsilon_{ijk} \epsilon_{ij'k'} = \delta_{jj'} \delta_{kk'} - \delta_{jk'} \delta_{j'k} \quad (4)$$

We get

$$F'_{\mu\nu}F^{\mu\nu'} = F_{\mu\nu}F^{\mu\nu} = 2\vec{B}^2 - 2\vec{E}^2 \quad (5)$$

We can arrive at the identity

$$\vec{E}^2 - \vec{B}^2 = \vec{E}'^2 - \vec{B}'^2 \quad (6)$$

For our second identity we have.

$$F_{\mu\nu} \tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \quad (7)$$

$$= \frac{1}{2} \epsilon^{0ijk} F_{0i} F_{jk} \quad (8)$$

$$F_{\mu\nu} \tilde{F}^{\mu\nu} = 4(\vec{E} \cdot \vec{B}) \quad (9)$$

Given that

$$F'_{\mu\nu} \tilde{F}'^{\mu\nu} = F_{\mu\nu} \tilde{F}^{\mu\nu} \quad (10)$$

Which gives us

$$\vec{E} \cdot \vec{B} = \vec{E}' \cdot \vec{B}' \quad (11)$$

(1) E field vanishes

If the electric field vanishes $\vec{E}' = 0$ then we can use (11) to give us

$$\vec{E} \cdot \vec{B} = 0 \quad (12)$$

So we restrict $\vec{E}$ and $\vec{B}$ to be orthogonal.

Using (7) we get

$$\vec{E}^2 - \vec{B}^2 = -\vec{B}'^2 \quad (13)$$

Thus we have the restriction $|\vec{B}| > |\vec{E}|$.

If we define an arbitrary

$$\Lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} \quad (14)$$

With the restrictions of $\forall \lambda_i \in \mathbb{R}$, $2\lambda_3 - 3\lambda_2 > 1$, $3\lambda_1 - \lambda_3 > 2$ and $\lambda_2 - 2\lambda_1 > 3$

We can now write the magnetic field in the reference system as

$$\vec{B} = \vec{E} \times \vec{\Lambda} \quad (15)$$

The three restrictions above ensure $|\vec{B}| > |\vec{E}|$ holds while the wedge ensures orthogonality.

(2) Magnetic field Vanishing

If the magnetic field vanishes $\vec{B}' = 0$ then we can use (11) again to give us

$$\vec{E} \cdot \vec{B} = 0 \quad (16)$$

So we restrict $\vec{E}$ and $\vec{B}$ to be orthogonal.
Using (7) we get

$$\vec{E}^2 - \vec{B}^2 = \vec{E}'^2 \quad (17)$$

Thus we get the inverse restriction $|\vec{B}| < |\vec{E}|$.

So we can reverse our inequalities and get the same result. With the restrictions of $\forall \lambda_i \in \mathbb{R}$, $2\lambda_3 - 3\lambda_2 < 1$, $3\lambda_1 - \lambda_3 < 2$ and $\lambda_2 - 2\lambda_1 < 3$

We can now write the magnetic field in the reference system as

$$\vec{B} = \vec{E} \times \vec{\Lambda} \quad (18)$$

The three restrictions above ensure the opposite restriction $|\vec{B}| < |\vec{E}|$ holds, while the wedge ensures orthogonality again.

(3) $\vec{E}$ and $\vec{B}$ are parallel

If they are parallel then 11 tells us nothing so we can pick an arbitrary $\vec{B}$ as long as it is not perpendicular to $\vec{E}$.

(4) $\vec{E}$ and $\vec{B}$ orthogonal

From (11) we have

$$\vec{E} \cdot \vec{B} = 0 \quad (19)$$

So if we let

$$\vec{B} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \quad (20)$$

Then we have any $\vec{B}$ satisfying

$$b_1 + 2b_2 + 3b_3 = 0 \quad (21)$$

for example if we had $b_1 = b_2 = 1$ and $b_3 = -1$

Question 2

Given

$$\Lambda = e^{-\zeta K_3} = \sum_{n=0}^{\infty} \frac{(-\zeta K_3)^n}{n!} \quad (22)$$

We can look at a few powers of K_3 and try to find a pattern.

I will notate the γ, δ th entry by $K_3^{\gamma\delta}$ with $\gamma, \delta \in \{0, 1, 2, 3\}$

$$K_3 = K_3^1$$

Computing K_3^2 we get

$$K_3^{00} = K_3^{33} = 1 \text{ otherwise } K_3^{\gamma\delta} = 0 \quad (23)$$

Computing K_3^3 we get

$$K_3^3 = K_3^1 = K_3 \quad (24)$$

Computing K_3^4 we get (yes i did this by hand)

$K_3^{00} = K_3^{33} = 1$ otherwise $K_3^{\gamma\delta} = 0$ Which is exactly K_3^2 (25) We could prove this using induction or we could be observant physicists and generalise to see that for n odd we have just K_3 while for n even we get an empty matrix with a 1 in the top left and bottom right. Let me call this matrix M . Thus in the expansion of Λ we can examine the even terms and the odd terms.

First for the even terms we have

$$E = \sum_{n=0}^{\infty} \frac{(-\zeta K_3)^{2n}}{(2n)!} \quad (26)$$

Note: A matrix to the power 0 gives an identity term $I_{4 \times 4}$ here. We can decompose this identity into the two following matrices.

$$I_{t,z} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ and } I_{x,y} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (27)$$

We can identify

$$\cosh(x) = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!(28)}$$

So we have

$$E = \cosh(\zeta) \cdot M + I_{x,y} \quad (29)$$

Similarly for odd terms we have

$$O = \sum_{n=0}^{\infty} \frac{(-\zeta K_3)^{2n+1}}{(2n+1)!} \quad (30)$$

We can identify

$$\sinh(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!(31)}$$

So we have

$$O = -\sinh(\zeta) \cdot K_3 \quad (32)$$

Combining these we get

$$\Lambda = \cosh(\zeta)M - \sinh(\zeta)K_3 + I_{x,y} \quad (33)$$

In matrix form this looks like

$$\Lambda = \begin{bmatrix} \cosh(\zeta) & 0 & 0 & -\sinh(\zeta) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh(\zeta) & 0 & 0 & \cosh(\zeta) \end{bmatrix} \quad (34)$$

Given the that we have $\gamma = \cosh(\zeta)$ and $\sinh(\zeta) = \gamma\beta$. We get

$$\Lambda = \begin{bmatrix} \gamma & 0 & 0 & -\gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma\beta & 0 & 0 & \gamma \end{bmatrix} \quad (35)$$

For the Velocity of O' as measured in O we can see how the coordinates transform. Given $\beta = \tanh(\zeta)$. We have the velocity of O' relative to O to be give by

$$x'^{\mu} = \Lambda_{\nu}^{\mu} x^{\nu} \quad (36)$$

If we expand the last row ($\mu = 3$) we get

$$z' = -\gamma\beta ct + (\gamma)z \quad (37)$$

Taking the origin of O' we get $z = \beta ct$ with $v = \frac{dz}{dt}$. So we have

$$v = \frac{d}{dt}(\beta ct) \quad (38)$$

$$v = \beta c = c \tanh(\zeta) \quad (39)$$

$$F^{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{bmatrix} \quad (40)$$

Under transformations we compute

$$F'^{\mu\nu} = \Lambda_{\alpha}^{\mu} \Lambda_{\beta}^{\nu} F^{\alpha\beta} \quad (41)$$

In matrix form we have

$$F'^{\mu\nu} = \begin{bmatrix} \gamma & 0 & 0 & -\gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma\beta & 0 & 0 & \gamma \end{bmatrix} \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{bmatrix} \begin{bmatrix} \gamma & 0 & 0 & -\gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma\beta & 0 & 0 & \gamma \end{bmatrix}^T \quad (42)$$

$$F'^{\mu\nu} = \begin{bmatrix} -\frac{\beta E_z}{\sqrt{1-\beta^2}} & \frac{\beta B_y}{\sqrt{1-\beta^2}} - \frac{E_x}{\sqrt{1-\beta^2}} & -\frac{\beta B_x}{\sqrt{1-\beta^2}} - \frac{E_y}{\sqrt{1-\beta^2}} & -\frac{E_z}{\sqrt{1-\beta^2}} \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ \frac{E_z}{\sqrt{1-\beta^2}} & \frac{\beta E_z}{\sqrt{1-\beta^2}} - \frac{B_y}{\sqrt{1-\beta^2}} & \frac{B_x}{\sqrt{1-\beta^2}} + \frac{b E_y}{\sqrt{1-\beta^2}} & \frac{\beta E_z}{\sqrt{1-\beta^2}} \end{bmatrix} \begin{bmatrix} \gamma & 0 & 0 & -\gamma\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma\beta & 0 & 0 & \gamma \end{bmatrix} \quad (43)$$

$$F'^{\mu\nu} = \begin{bmatrix} 0 & \frac{\beta B_y}{\sqrt{1-\beta^2}} - \frac{E_x}{\sqrt{1-\beta^2}} & -\frac{\beta B_x}{\sqrt{1-\beta^2}} - \frac{E_y}{\sqrt{1-\beta^2}} & \frac{\beta^2 E_z}{1-\beta^2} - \frac{E_z}{1-\beta^2} \\ \frac{E_x}{\sqrt{1-\beta^2}} - \frac{\beta B_y}{\sqrt{1-\beta^2}} & 0 & -B_z & \frac{B_y}{\sqrt{1-\beta^2}} - \frac{\beta E_x}{\sqrt{1-\beta^2}} \\ \frac{\beta B_x}{\sqrt{1-\beta^2}} + \frac{E_y}{\sqrt{1-\beta^2}} & B_z & 0 & -\frac{B_x}{\sqrt{1-\beta^2}} - \frac{\beta E_y}{\sqrt{1-\beta^2}} \\ \frac{E_z}{1-\beta^2} - \frac{\beta^2 E_z}{1-\beta^2} & \frac{\beta E_x}{\sqrt{1-\beta^2}} - \frac{B_y}{\sqrt{1-\beta^2}} & \frac{B_x}{\sqrt{1-\beta^2}} + \frac{\beta E_y}{\sqrt{1-\beta^2}} & 0 \end{bmatrix} \quad (44)$$

$$F'^{\mu\nu} = \begin{bmatrix} 0 & \gamma (\beta B_y - E_x) B_y & -\gamma (\beta B_x + E_y) & -E_z \\ \gamma (E_x - \beta B_y) & 0 & -B_z & \gamma (B_y - \beta E_x) \\ \gamma (\beta B_x + E_y) & B_z & 0 & -\gamma (B_x + \beta E_y) \\ E_z & \gamma (\beta E_x - B_y) & \gamma (B_x + \beta E_y) & 0 \end{bmatrix} \quad (45)$$

We can read off:

$$E'_x = \gamma (E_x - \beta B_y) \quad (46)$$

$$E'_y = \gamma (\beta B_x + E_y) \quad (47)$$

$$E'_z = E_z \quad (48)$$

$$B'_x = \gamma (B_x + \beta E_y) \quad (49)$$

$$B'_y = \gamma (\beta E_x - B_y) \quad (50)$$

$$B'_z = B_z \quad (51)$$

(3) Rotation about z-axis

To define our lorentz boost for a rotation about the z axis we need to have rotation matrix in the $x - y$ plane. This boost is given by

$$\Lambda_{\nu}^{\mu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha & 0 \\ 0 & \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (52)$$

Then we perform a transformation.

$$F'^{\mu\nu} = \Lambda_{\alpha}^{\mu} \Lambda_{\beta}^{\nu} F^{\alpha\beta} \quad (53)$$

Since the rotation is in the $x - y$ plane the only non-invariant components are the x and y components. We can apply our rotation matrix

$$\begin{bmatrix} E'_x \\ E'_y \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} E_x \\ E_y \end{bmatrix} \quad (54)$$

We recieve

$$E'_x = E_x \cos \alpha - E_y \sin \alpha \quad (55)$$

$$E'_y = E_x \sin \alpha + E_y \cos \alpha \quad (56)$$

We can interchange E_{ρ} and B_{ρ} to find out how the magnetic field transforms. We obtain

$$B'_x = B_x \cos \alpha - B_y \sin \alpha \quad (57)$$

$$B'_y = B_x \sin \alpha + B_y \cos \alpha \quad (58)$$

Question 3

We have the in homogeneous wave equation for $A^\mu(x)$ given by

$$\square A_\mu(x) = \frac{4\pi}{c} j_\mu(x) \quad (59)$$

with

$$\square D_{ret}(x) = \delta^4(x) \quad (60)$$

$$A_\mu(x) = \int d^4x' D_{ret}(x-x') \frac{4\pi}{c} j_\mu(x') \quad (61)$$

$$A_\mu(x) = \int d^4x' \frac{1}{2\pi} \theta_H(x_0 - x'_0) \delta((x-x')^2) \frac{4\pi}{c} j_\mu(x') \quad (62)$$

$$A_\mu(x) = \int d^4x' \theta_H(x_0 - x'_0) \delta((x-x')^2) \frac{2}{c} j_\mu(x') \quad (63)$$

The heavyside function ensures that the only non-vanishing contribution is for $x_0 \geq x'_0$. (retarded sol)

$$\text{with } j_\mu \left(\frac{x_0 - \frac{|x-x'|}{c}, x'}{|x-x'|}, v \right), v = D_{ret}(x-x').$$

For the Lorentz condition we have:

$$\partial^\mu A_\mu(x) = \frac{4\pi}{c} \int d^4x' [\partial^\mu D_{ret}(x-x')] j_\mu(x') \quad (64)$$

Using IBP with

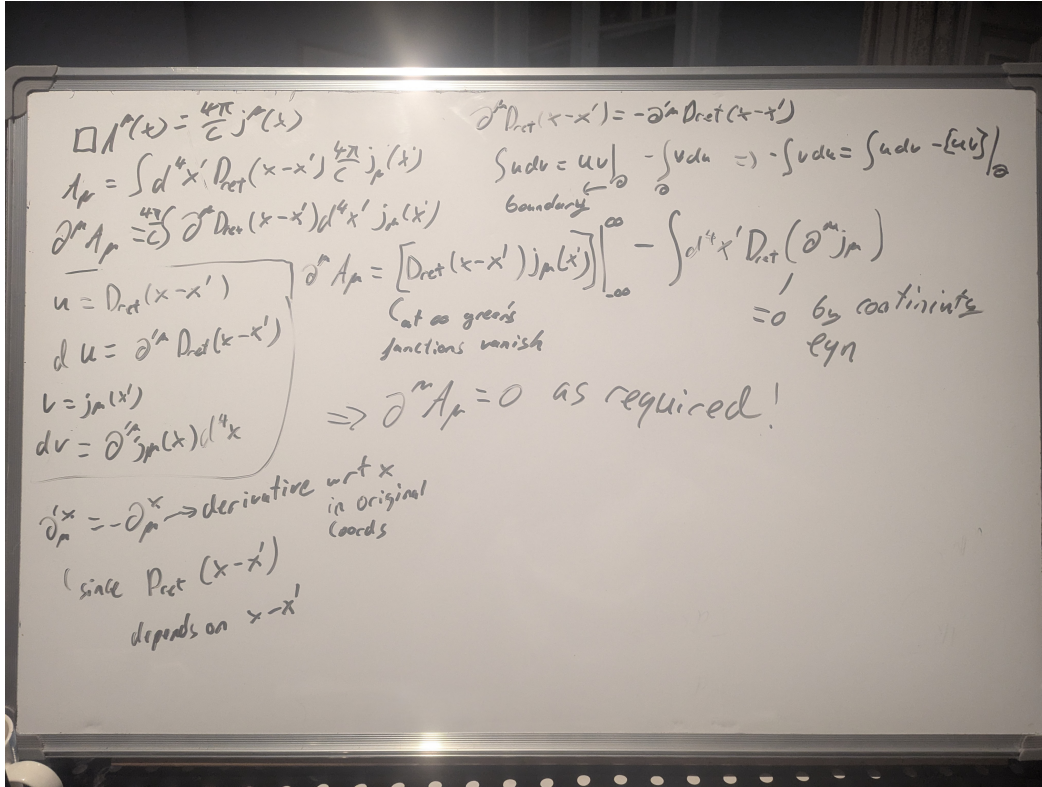


Figure 1: Sorry I am sending a picture of my whiteboard. I tried to type this out on latex but for some reason by environment throws errors when i have mu's as subscripts for some reason and i couldn't debug it on time.

3.2

If $A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \chi$

We know this must satisfy $\partial^\mu A_\mu = 0$ Thus we have $\partial^\mu A_\mu + \partial^\mu \partial_\mu \chi = 0$
Thus we have

$$\partial^\mu \partial_\mu \chi = -\partial^\mu A_\mu (x) \quad (65)$$

Which is just the inhomogeneous wave equation XXX INCLUDE STEPS We get

$$\chi(x) = - \int d^4 x' D_{ret}(x - x') (\partial^\mu A_\mu(x')) \quad (66)$$

$$\chi(x) = -1 \frac{1}{4\pi} \int \frac{\partial^\mu A_\mu(t - \frac{|x-x'|}{c}, x')}{|x - x'|} \quad (67)$$

Question 4

We know that for a point charge we have

$$\phi = \frac{1}{4\pi\epsilon_0} q \frac{1}{\sqrt{x^2 + y^2 + z^2}} (\hat{x} + \hat{y} + \hat{z}) \quad (68)$$

$$E = -\nabla\phi \quad (69)$$

$$\vec{E}_x = \frac{1}{4\pi\epsilon_0} q \frac{-x}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \quad (70)$$

$$\vec{E}_y = \frac{1}{4\pi\epsilon_0} q \frac{-y}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \quad (71)$$

$$\vec{E}_z = \frac{1}{4\pi\epsilon_0} q \frac{-z}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \quad (72)$$

$\vec{B} = 0$ as we have a point charge which isn't moving.

We can read off the components of the field tensor

$$F'^{\mu\nu} = \Lambda_\alpha^\mu \Lambda_\beta^\nu F^{\alpha\beta} \quad (73)$$

$$F'^{10} = \Lambda_\alpha^1 \Lambda_\beta^0 F^{\alpha\beta} = \Lambda_1^1 \Lambda_0^0 F^{10} + \Lambda_0^1 \Lambda_1^0 F^{01} = \gamma^2 - E_x + \gamma^2 \beta^2 (E_x) = -\gamma^2 E_x (1 - \beta^2) = -E'_x \quad (74)$$

$$F'^{20} = E'_y = \Lambda_\alpha^2 \Lambda_\beta^0 F^{\alpha\beta} = \Lambda_2^2 \Lambda_0^0 F^{20} + \Lambda_2^0 \Lambda_1^0 F^{21} = -\gamma E_y + \gamma \beta B_z = E'_y \quad (75)$$

$$F'^{30} = E'_z = \Lambda_\alpha^3 \Lambda_\beta^0 F^{\alpha\beta} = \Lambda_3^3 \Lambda_0^0 F^{30} + \Lambda_3^0 \Lambda_1^0 F^{31} = -\gamma E_z + -\gamma (\beta) (-B_y) = E'_z \quad (76)$$

$$F'^{32} = B'_x = \Lambda_\alpha^3 \Lambda_\beta^2 F^{\alpha\beta} = \Lambda_3^3 \Lambda_2^2 F^{32} = B_x \quad (77)$$

$$F'^{31} = -B'_y = \Lambda_\alpha^3 \Lambda_\beta^1 F^{\alpha\beta} = \Lambda_3^3 \Lambda_0^1 F^{30} + \Lambda_3^0 \Lambda_1^1 F^{31} = -\gamma \beta E_z - \gamma B_y = -B'_y \quad (78)$$

$$F'^{21} = B'_z = \Lambda_\alpha^2 \Lambda_\beta^1 F^{\alpha\beta} = \Lambda_2^2 \Lambda_0^1 F^{20} + \Lambda_2^0 \Lambda_1^1 F^{21} = -\gamma \beta E_y + \gamma B_z = B'_z \quad (79)$$

Note above I read off $F^{\nu\mu}$ by mistake but below I multiply the results by a minus sign to fix this error due to the fact that F is anti symmetric namely $F^{\mu\nu} = -F^{\nu\mu}$. Thus explicitly we have

$$\vec{E}' = [[E_x], [\gamma(E_y - \beta B_z)], [\gamma(E_z + \beta B_y)]] \quad (80)$$

$$\vec{B}' = [[B_x], [\gamma(\beta E_z + B_y)], [-\gamma(B_z - \beta E_y)]] \quad (81)$$

Since in the original frame we have no magnetic field these reduce to

$$\vec{E}' = [[E_x], [\gamma(E_y)], [\gamma(E_z)]] \quad (82)$$

$$\vec{B}' = [[0], [-\gamma(\beta E_z)], [\gamma(\beta E_y)]] \quad (83)$$

For the lorentze invariants we have

$$\vec{E}' \cdot \vec{B}' = \gamma^2 \beta E_y E_z - \gamma^2 \beta E_y E_z = 0 \quad (84)$$

Which matches $\vec{B} \cdot \vec{E} = 0$ since $\vec{B} = 0$.
Likewise for $\vec{E}^2 - \vec{B}^2 = \vec{E}^2$.

$$\vec{E'}^2 - \vec{B'}^2 = E_x^2 + \gamma^2 E_y^2 + \gamma^2 E_z^2 - \gamma^2 \beta^2 E_z^2 - \gamma^2 \beta^2 E_y^2 \quad (85)$$

$$\vec{E'}^2 - \vec{B'}^2 = E_x^2 + \gamma^2 (1 - \beta^2) E_y^2 + \gamma^2 (1 - \beta^2) E_z^2 \quad (86)$$

$$\vec{E'}^2 - \vec{B'}^2 = E_x^2 + E_y^2 + E_z^2 = \vec{E}^2 \quad (87)$$